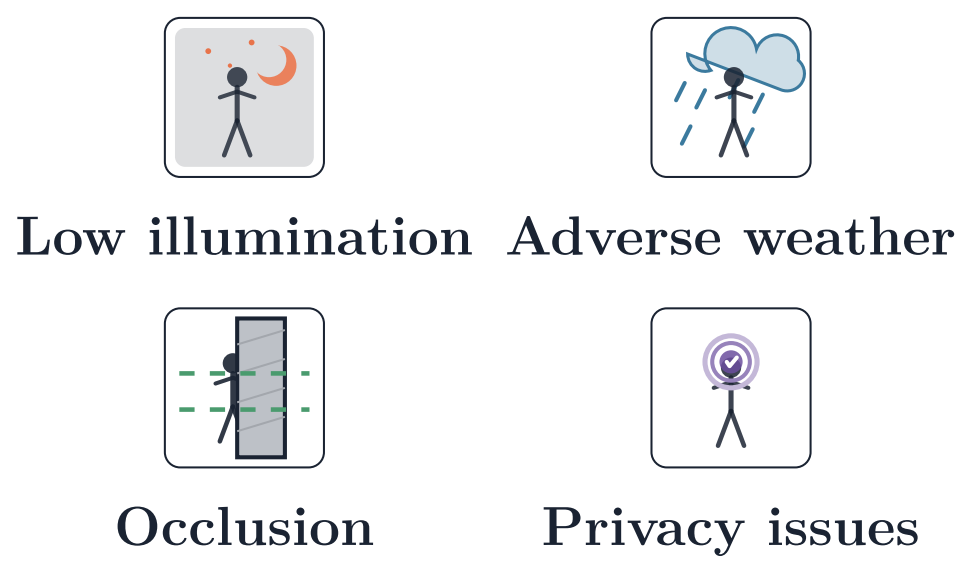


# Few-Shot Radar Person Identification via Tracking-Integrated InISAR and Pre-Trained Meta-Learning

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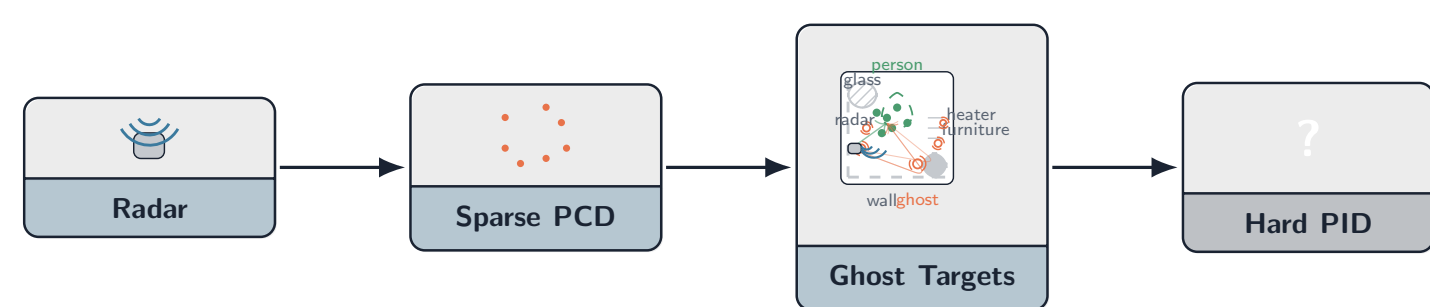
## 1 Introduction

Vision-based person identification (PID) struggle under conditions [1, 2]:

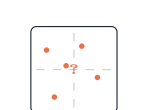


**Radar Advantages:** Zero optical exposure · All-weather operation · Penetrates thin barriers · Privacy preserving

## 2 Challenges in Radar-Based PID



Sparse, noisy point clouds with multipath ghosts.



### Low Angular Resolution

Conventional angle-of-arrival [3] yields motion-defocused clouds and multipath artifacts.



### Rigid-Target Bias

Interferometric Inverse Synthetic Aperture Radar (InISAR) improves resolution but is validated only on large *rigid* targets [4, 5].



### Multipath Ghosts

Ghosts overlap in range-Doppler and mimic real 3D clusters [6].



### Data Scarcity

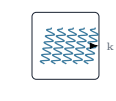
Existing SOTA [7, 8] demand > 100 seq/person; Point Cloud Transformer (PCT) [9] handles spatial structure but targets *static* objects [10].

**Gap:** Tracking-integrated InISAR + pre-trained meta-learning.

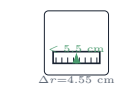
## 3 Key Results

**73.8%** **< 5.5 cm**  
InISAR IoU Height Error  
**70.0%**  
Cross-Session PID

### Contributions & System Impact



**Articulated Validation:** First InISAR validated on humans in multipath-rich indoor envs.



**Resolution-level Error:** < 5.5 cm (med. 3.87 cm) approaches  $\Delta r=4.55$  cm limit (vs. prior 0.2–1.6 m errors).



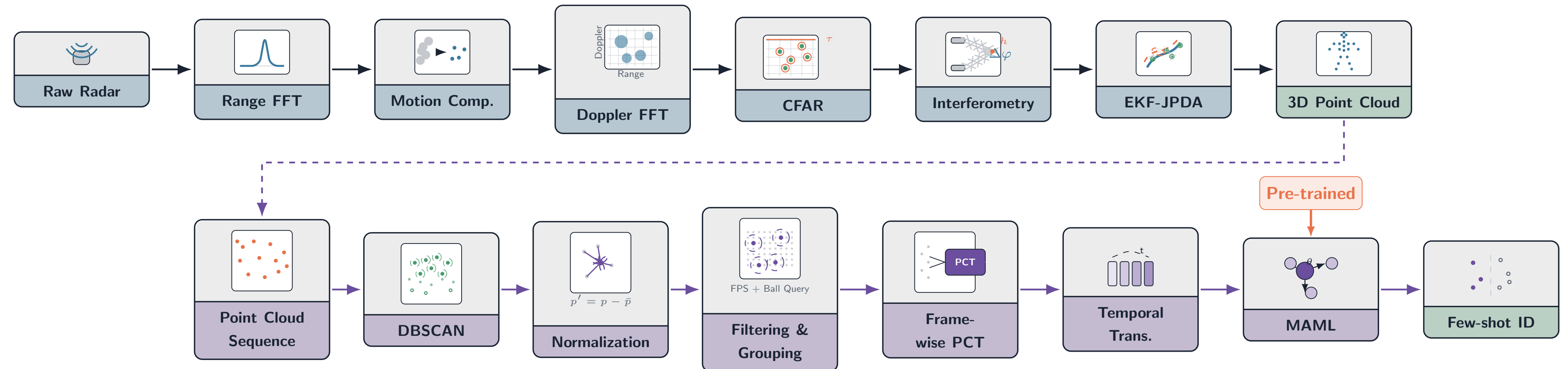
**SOTA Supervised:** PCT-Trans achieves 98.99% supervised (+2.81pp over TPCN).



**MAML Gains:** Pre-trained MAML yields 70.0% cross-session (+15.6pp over random).

**Deployable few-shot radar PID with minimal training data.**

## 4 Our Framework: Unified Sensing + Meta-Learning



End-to-end system: InISAR 3D reconstruction → spatio-temporal learning → PID

### (1) Radar domain

Tracking-integrated InISAR reconstructs high-res 3D point clouds of articulated human motion, validated against motion-capture ground truth.

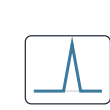
### (2) ML domain

Spatio-temporal transformer (PCTrans) embedded in Model-Agnostic Meta-Learning (MAML); task-specific pre-trained initialization substantially improves few-shot adaptation.

### (3) System integration

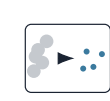
High-quality 3D reconstruction + meta-learning together enable practical few-shot radar PID, reducing dependence on large subject-specific datasets.

## 5 InISAR Pipeline: 3D Reconstruction



### 1. Range FFT

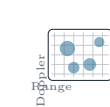
- FMCW chirp compression to  $S_{R,m}(f, t)$  per  $m$ -th antenna.
- Physical range limit  $\Delta r = c/(2B)$  bounds achieved error.



### 2. Motion Compensation (ICBA)

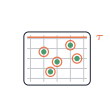
- Translational model  $R_0(t) = r_0 + \beta t + \gamma t^2$ .
- Nelder-Mead optimizes contrast  $[\hat{\beta}, \hat{\gamma}]$  to remove smearing → yields *signal-to-noise ratio (SNR) enhanced*  $S_{C,m}(f, t)$ :

$$[\hat{\beta}, \hat{\gamma}] = \arg \max \left\{ \frac{\sqrt{\frac{1}{N} \sum (I_i - \bar{I})^2}}{I} \right\}$$



### 3. Coherent Combination & Doppler FFT

- Geometric phase corrections align all MIMO virtual antennas (SNR boost) → *focused* 2D Range-Doppler Profile (RDP).



### 4. CFAR Segmentation

- SOCA-CFAR ( $N_T = 6$ ,  $P_{FA} = 10^{-10}$ ) isolates target cells from noise/clutter.



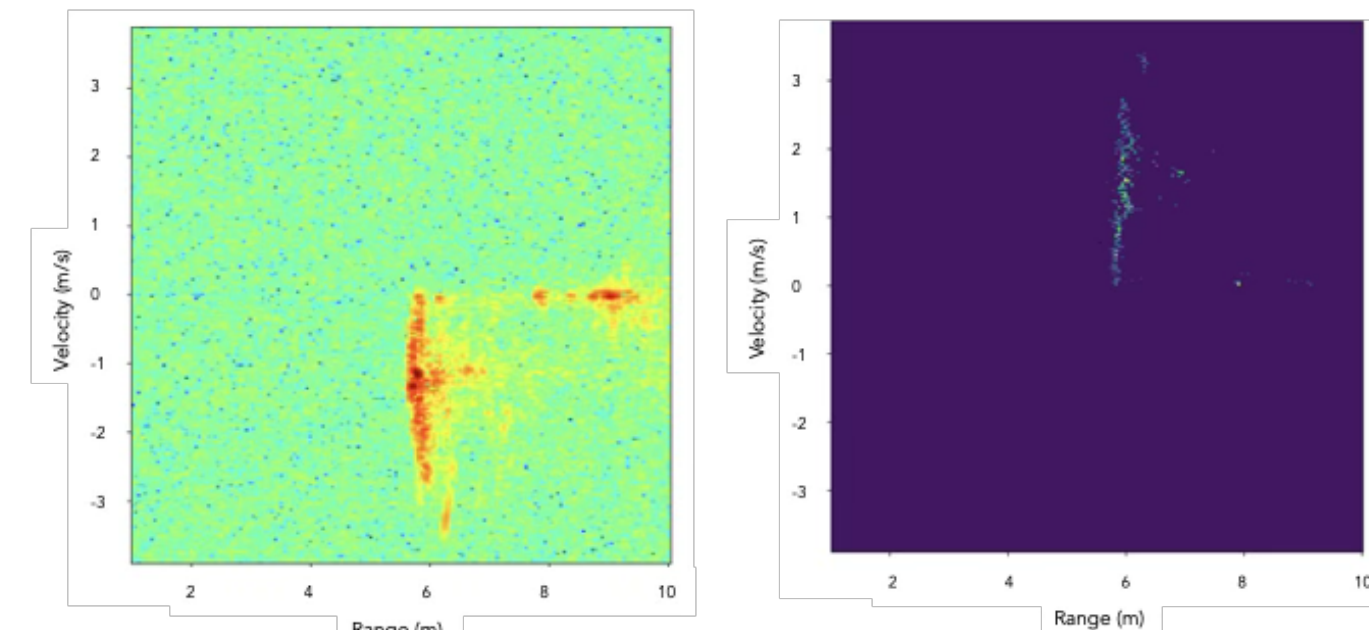
### 5. Interferometry

- Phase difference between top/bottom MIMO rows *yields elevation*:  
$$I_1 \cdot I_2^* = |I_1| |I_2| \exp(j \frac{2\pi}{\lambda} \Delta R) \rightarrow h = \frac{\lambda \phi R}{2\pi L}$$
- Exploiting MIMO antenna pairs reduces measurement noise.



### 6. EKF-JPDA Tracking

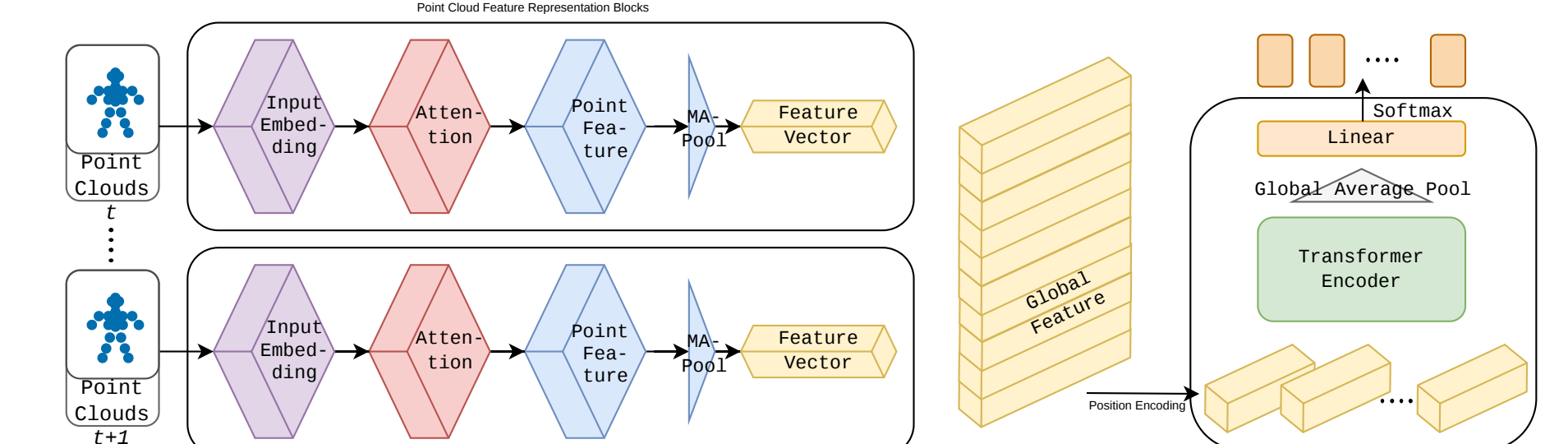
- Spatial-domain JPDA handles data association; EKF filters state covariance.
- Max miss count=1 → *prunes multipath ghosts* while tolerating occlusions.



Raw RDP

Post-InISAR RDP

## 6 Few-Shot PID with PCTrans



PCTrans: Shared Spatial Extractor + Temporal Gait Model



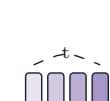
### Gait-Invariant Processing

- Centroid norm.  $p'_i = p_i - (\bar{x}, \bar{y}, 0)$  prevents path overfitting



### Spatial PCT Encoder

- Farthest Point Sampling → non-uniform density
- Ball Query → rotation-invariant neighborhoods.
- Neighbor Embedding:  $\Delta F(p) = \text{concat}_{q \in \text{km}(p,k)} (F(q) - F(p))$
- Offset-Attention acts as a discrete Laplacian:  
$$F_{\text{out}} = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}} \quad (f_t \in \mathbb{R}^{192})$$



### Temporal Transformer

- Positional encoding injects order
- Multi-head self-attention captures cross-frame gait dynamics:  $p(y|S) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1 \mathbf{g} + b_1) + b_2)$

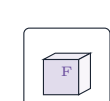
## 7 Pre-trained Meta-Learning [11]



### Two-Phase Domain-Aware Adaptation

- Phase 1:** Supervised pre-training (3 subjects) →  $\theta_0^{PT}$  (session-invariant patterns).
- Phase 2:** MAML meta-training [12, 13].

Inner:  $\theta'_i = \theta - \alpha \nabla \mathcal{L}$  Outer:  $\theta \leftarrow \theta - \beta \nabla \sum_i \mathcal{L}$

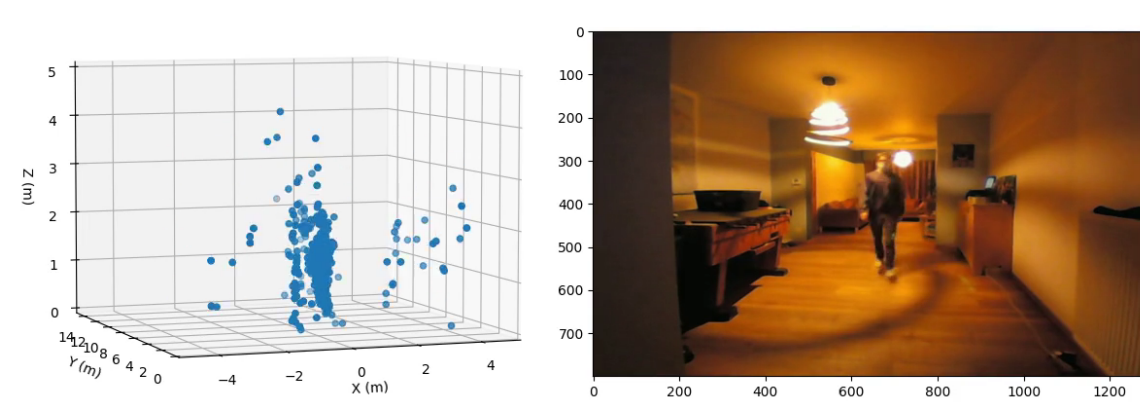


**Key Insight:**  $\theta_0^{PT}$  lies closer to optimal  $\theta^*$ , enabling faster convergence than random initialization.

**Gain:** +15.6pp over random MAML

## 8 Experimental Results

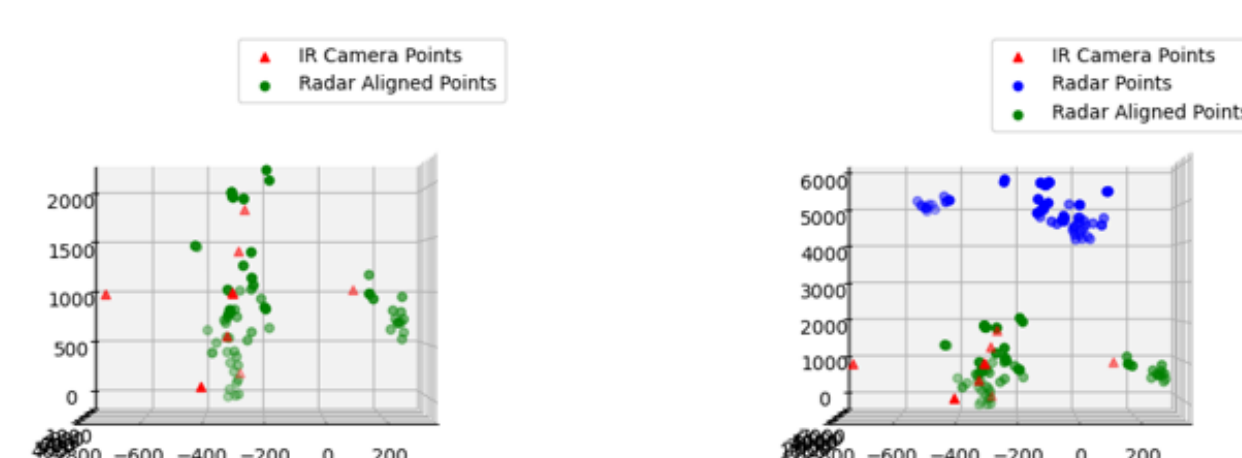
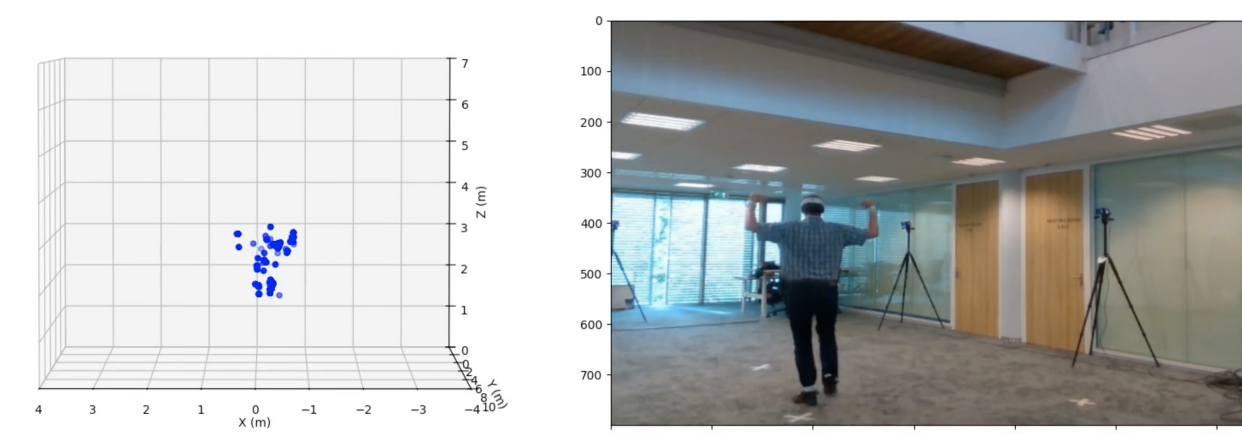
### Ghost Removal (EKF-JPDA)



(a) Before: Ghost remains

(b) After: Ghost rejected

### InISAR vs Motion Capture



Reconstruction Height Error < 5.5 cm (median 3.87 cm)

| Scenario         | IoU (%) |
|------------------|---------|
| Circular Walking | 79.82   |
| Hands-Up Walking | 100.0   |
| Kicking          | 69.19   |
| Squatting        | 59.98   |
| Average          | 73.78   |

**Dataset:** 5 subj. (164–185 cm),  $B = 3.3$  GHz ( $\Delta r = 4.55$  cm).

### PID Benchmark Accuracy (%)

| Query Type        | TCPCN [8] | PCT-T MAML (sup.) | MAML (rand.) | MAML (pre-train) |
|-------------------|-----------|-------------------|--------------|------------------|
| Q1(same session)  | 96.18     | 98.99             | 76.67        | 94.44            |
| Q2(cross-session) | 35.10     | 32.05             | 54.44        | 70.00            |
| Q3(cross-subject) | 57.81     | 40.09             | 46.67        | 48.89            |

- Supervised:** PCT-T outperforms TCPCN → +2.81pp.
- Few-Shot:** Pre-trained MAML vs Random → +15.56pp.
- mmGait Benchmark:** PCT-T leads by +11.5pp on fixed trajectories [14].